

The umbilical connects to the main Hi-FLi unit via a male 8-pin Cinch-Jones plug. This carries $\pm 12\text{V}$ power into the pedal unit preamp board and control voltages, bypass switching voltage and audio into the main unit.

Hi-FLi Main Unit Overview

Looking at the Hi-FLi main console we see a collection of effects, each of which has a slider below it and below each of these (with the exception of the main mix slider) there is a 3 way slide switch. These slide switches engage control voltages from the pedal units. Indeed all the effects in the Hi-FLi are voltage controllable, a very radical concept in 1972 and one which gives the Hi-FLi huge power and versatility to create many complex and diverse effects.

Here is a brief summary of the main effects (starting left to right). A more detailed description of each effect and its structure will be given later.

INPUT SIGNAL SENSITIVITY

Solo/Strum Switch

Part of the Hi-FLi circuitry is to detect the start and end of the amplitude envelope of audio input signals, e.g. from a plucked guitar string(s). Two pulses are created one at the beginning (the 'attack' phase) of the amplitude envelope and another pulse at the end (the 'decay' phase) of the envelope.

These pulses are important in triggering/re-triggering the various dynamical modulation waveforms used in the Hi-FLi effects during playing of e.g. a guitar. The solo/strum settings refer to the sensitivities of this 'attack' and 'decay' detector circuit. The solo setting is the more sensitive and is designed to create the trigger/retrigger pulses when playing single strings. The strum setting is less sensitive and is designed to obtain correct trigger/retrigger pulses when playing chords where the amplitude envelope is both larger and more complex in shape than for single plucked strings.

A more detailed discussion of this will be given later.

TOP BOOST

Audio in from the preamp in the pedal unit is fed to the Top Boost that is basically a treble boost. It gives up to 30db boost of higher frequencies compensated to keep overall sound levels constant. It's a very useful feature that adds dramatic tonal qualities to the sound particularly in combination with the other Hi-FLi effects.

OCTAVE SHIFT

Sub Octave

This section involves a sub-octave generator that extracts a fundamental note from the audio input and drops it by one octave. Raising the fader increases the mix of dry and octave-shifted signal.

Buzz Switch

The sub-octave signal is a square wave. The buzz switch allows the option of converting this to a triangle wave which adds different harmonics providing a more 'buzzy/raspy' sound effect.

Ring Mod

The ring modulator effect in the Hi-FLi is quite different in design from typical ring mod circuits found in synthesizers. The latter basically multiply two wave forms together, so typically this results in a doubling of frequency if two pure tones are fed into it plus many other harmonics, giving rise to bell like ringing and metallic sounds.

In the Hi-FLi, a 'pseudo' ring modulation effect occurs where instead of multiplying two waveforms, the ring mod just adds a half wave rectified version of the fundamental to the output of the Top Boost and thus can produce a fundamental that is double in frequency. This 'pseudo ring mod' signal is sent to a pair of FET transistors controlled by the sub-octave signal itself. Because of this one doesn't hear metallic/bell like sounds. Rather the result is some great deep sub-octaves (with complex harmonics if the signal in is not just a pure tone) the amount of which are controlled by the ring-mod and sub-octave facers. So basically the sub-octave and ring mod effects are interconnected in a fairly complicated way.

Decay Rate

This controls the decay time of the audio input as it passes through top-boost, sub-octave and ring-mod effects. The Hi-FLi circuits create an envelope follower of the input signal amplitude and this envelope controls the level of the signal passing through these 3 effects. The decay rate knob controls the decay rate of the envelope. On fast setting the envelope will be very short, leading to clipped like plucks that 'sample' the initial input signal attack. On longer decay settings the envelope will act more as a follower. In this way you have control over the input signal envelope dynamics as it passes through effects further down the 'chain'.

SUSTAIN FUZZ

Attack Rate

Sustain fuzz is a silicon transistor based fuzz effect that incorporates some very clever and unique features to the Hi-FLi. The fuzz input signal is taken from the Top Boost section. The Hi-FLi detects the start and end of e.g. a plucked string from the envelope follower circuit mentioned above. Setting the fuzz attack rate fast, the fuzz appears instantly a string is plucked and the level controlled by the fader setting. But setting the time to slower values allows the fuzz to 'swell' in amplitude after the string is plucked. So you get a slewed or 'delayed' fuzz effect.

BYPASS MIXER

The main mix fader controls the amount of mix between 'dry' and 'wet' signals. At lowest setting 0, the signal is completely dry and the output of the pre-amplifier in the pedal unit goes directly to the audio output jack. At the maximum setting of 10, the signal route goes completely through all the effects in the Hi-FLi. Intermediate slider settings allow control over dry/wet mix. The bypass footswitch (which is built into the left hand pedal), switches between a fully dry signal and the mixed signal determined by the main fader setting.

Envelope & Fuzz Bypass Switch

This switch has two settings. In the 'normal' mode audio into the Hi-FLi goes through the standard routing discussed above. That is, the signal passes through Top Boost, Octave Shift and Sustain Fuzz sections. In the envelope and fuzz bypass mode, the audio input bypasses completely these three effects and enters unchanged into the Phase Filter section which will be described below. Thus none of the corresponding faders or the envelope decay rate/attack times knobs have any effect on the sound. The idea is to provide totally unmodified audio to the Phase Filter so that it can be used just by itself. This bypass switch was added to some later Hi-FLi's produced by E.M.S. in the 1970's.

Left Pedal Switches

Below each of the 4 sliders controlling Top Boost, Sub-Octave, Ring Mod and Fuzz level is a 3 way slide switch marked '+', 'o' and '-'. In the centre position the switch is off but in the other two settings they engage the control voltage generated by operating the left footpedal. These voltages vary between approximately $\pm 5V$ as the foot pedal operates between its two extremities. The two options '+', and '-' allow variations from negative to positive control voltage or from positive to negative for a movement of the pedal.

Recall that all the effects in the Hi-FLi are voltage controlled. Indeed each control slider is producing a voltage to control an effect when it is moved. By engaging the left pedal control voltages, Top Boost, Sub-Octave, Ring Mod and Fuzz level can be remote-controlled by operating the left pedal. The corresponding sliders still work and so the effect is controlled by the sum of the slider voltage and pedal voltage. The slider settings basically give the effect setting when the pedal is in the horizontal position (as opposed to 'up' or 'down') when the control voltage it puts out is close to zero.

PHASE FILTER

Here is where the major 'sound-scaping' effects occur. The Phase Filter consists of 12 filters together with FET/bipolar transistor switches that alter the signal path through the filters of audio fed from the left-hand effects.

Each of filters is of the 'all-pass' but frequency dependent phase shift type. This means each filter leaves the amplitude of any given frequency wave

unchanged but phase shifts the wave by an amount that depends on the waves frequency. The amount of phase shift in each filter can be controlled by a voltage. Varying the phase by applying modulation voltages together with the different signal path topologies through the filter allow for an amazing number of different effects some familiar some totally alien! It's likely that many of these phasing effects are unique to the Hi-FLi, and cannot be reproduced with modern effects units...analogue or digital.

Control Modulation Selector Switch

Six modulation waveforms that control the phase-shifting of the filters are available via this rotary switch. The first two are slow/fast continuous sine waves. The remainder are triggered or re-triggered waveforms, where the trigger/re-trigger pulses are generated by the attack/decay of the audio input envelope (more details of this will be discussed below). These waveforms are increasing amplitude sine waves; decreasing amplitude sine waves; ramp up and ramp down.

Treatment Selector Switch

Six different Treatments, which correspond to different topologies of audio signal paths through the 12 phase filters, are available via this rotary switch. The resulting effects are labelled as Vibrato, Phasing 1, Phasing 2, Waa, Waw and Meow. A detailed description of each of these will be given later.

Modulation Speed

This slider controls the speed of the Sine wave (constant as well as increasing/decreasing versions) modulation waveforms.

Modulation Ramp Time

Controls the ramp times of the modulation waveforms with dynamical envelopes, increasing/decreasing amplitude Sine waves and up/down ramps. For the Sine waveforms it controls the time taken for a wave to increase to its maximum amplitude/decrease to zero. For the ramp waveforms it controls the time taken for the amplitude of the ramp to reach maximum/minimum levels.

Modulation Depth

This slider controls the depth of the modulation waveforms controlling the amount of phase shifting produced the filters.

Frequency Shift

This controls the DC bias applied to the filters. The centre position (5) corresponds to 0 Volt bias. Above and below this a positive/negative bias is applied to each filter which controls the amount of phase-shifting applied to a given frequency wave. This control has a major affect on the sounds produced by the filters as it effectively shifts the DC bias of the modulation waveforms if applied. As a result, any applied modulation is acting within different frequency regions of the filter and produces dramatically different effects.

Growl Switch

The 'Growl' switch option was a feature that appeared on later Hi-FLi's produced by E.M.S in the 1970's, often referred to as 'mk2' units.

The switch is a 3-way centre off slide type. The growl circuit takes the sub-octave square wave from the Octave Shift section and divides its frequency by 2 and 4, thus producing a wave that drops the fundamental by 2 or 3 octaves (!) depending on the switch position. Rather than simply mixing these sub-harmonics with the original signal something rather more sophisticated is implemented.

These sub-harmonics are in fact used to modulate the phase filter but only when the filter control modulations is set to one of the 4 waveforms involving Sine waves (thus operating the Growl switch has no affect when the up/down ramp modulations are selected). This gives amazing/bizarre new modulations! The amount of growl is controllable via the modulation depth slider and also if the increasing/decreasing sine modulation is selected, by the modulation ramp time.

Growl can produce probably some of the most alien sounds from the Hi-FLi! Certainly it can turn a plucked guitar string note into something more associated with a synthesizer. Combined with all the other effects and treatments it can give a mind boggling array of sounds just from a single plucked guitar string. If two or more strings are struck then many additional complex harmonics and modulations are created (recall the Growl modulation of the phase filter is not something 'static' of fixed frequency. It changes with the pitch of the fundamental frequency extracted from a note.) This explains

why plucking two or more strings at same time produces very complex modulations.

Right Pedal Switches

The right pedal switches have the some functions as described above for the left pedal switches, only now they are associated with control voltage generated by movements of the right pedal and are applied to Modulation Speed, Modulation Ramp Time, Modulation Depth and Frequency Shift respectively.

Leds

On the right hand side of the control panel you will have noticed two red leds that glow when the unit is powered up. These are there to offer a basic visualization of the various control modulation waveforms as they are applied to the Phase Filter. We will discuss the leds and how to interpret their behaviour later on when we look at using the Hi-FLi in practice.

Finally, to help visualize how all the above effects and controls are linked together, in Fig 4 on the next page is a diagram representing the audio signal path through the Hi-FLi.

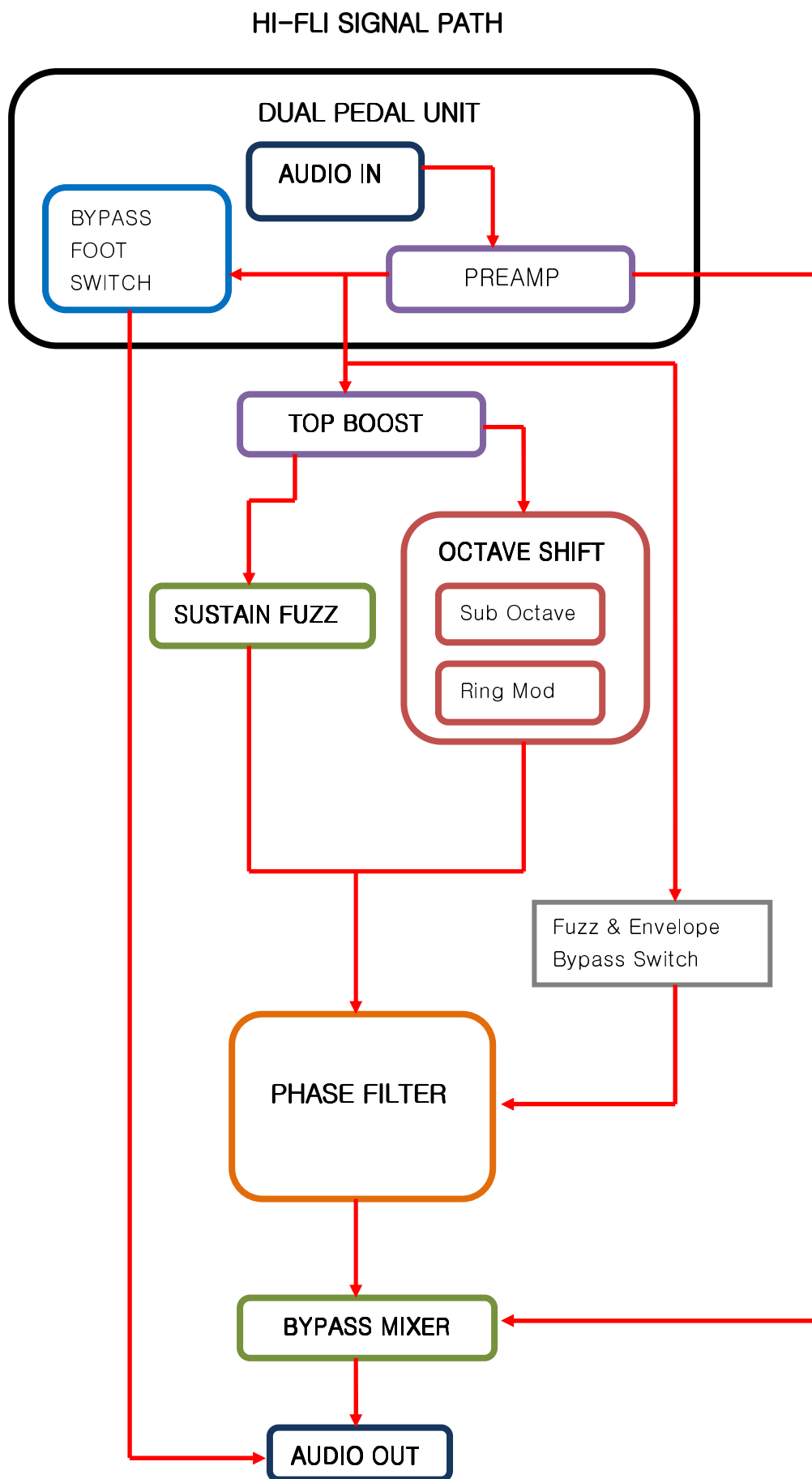


Fig 4

A Deeper Look into the Hi-FLi

In order to really get the most out of the Hi-FLi its helpful to understand in a little more detail two important aspects of its function, namely the Envelope Follower section and the various Phase Filter ‘treatments’ which were briefly described above.

Envelope Follower and the Attack/Decay Detector

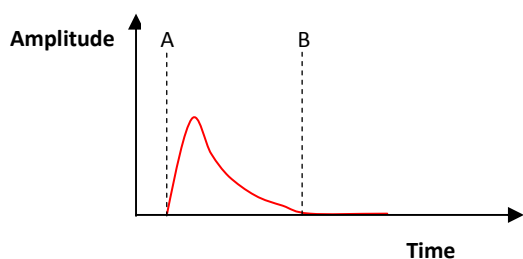


Fig 5a

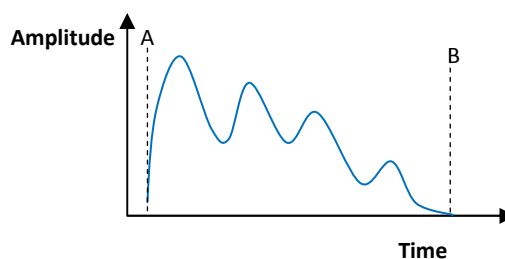


Fig 5b

Figs 5a and 5b illustrate a typical amplitude envelope when a single string is plucked (Fig 5a) compared to the case where several strings are strummed (Fig 5b). In the first plot we see a fast attack typical of a single plucked string quickly reaching a peak and then decaying in time. In the second plot again there is again a fast attack but the overall amplitude is larger and typically there may be several peaks in the amplitude before decaying, depending on how the chord is sounded.

In order to trigger and re-trigger some of the modulation waveforms in the Phase Filter section, the Hi-FLi needs to work out the start time and end time of the amplitude envelope, marked at A and B in the plots. This circuit is called the ‘attack/decay’ detector inside the Hi-FLi. Its design is certainly very clever, but the problem is that the second envelope has multiple peaks in it that can confuse the Hi-FLi into thinking that a single string has been plucked multiple times when in fact a single chord has been strummed once! The Hi-FLi generates a voltage pulse to mark the start and end of an amplitude envelope

which is then used to trigger/retrigger modulating waveforms applied to the Phase Filter as well as controlling the fuzz swell retriggering.

So without making any compensation for these two kinds of envelopes, the Hi-FLi might produce the pulses illustrated in Figs 6a and 6b

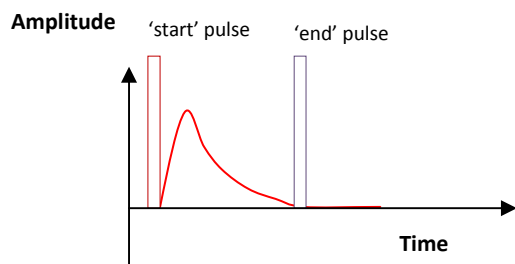


Fig 6a

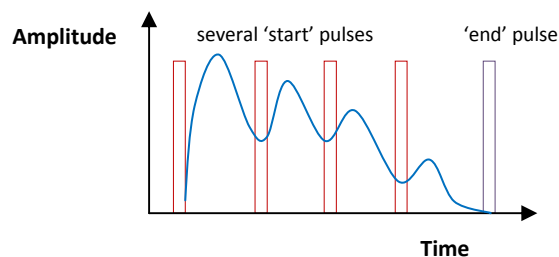


Fig 6b

In Fig 6a the Hi-FLi correctly generates a single pulse at the start and end of the envelope of a single plucked string but the same circuit when fed with the envelope of a chord strummed once as in Fig several start pulses!

The modulating 'one-shot' waveforms that are triggered/retriggered by these pulses are shown in Fig 7 below

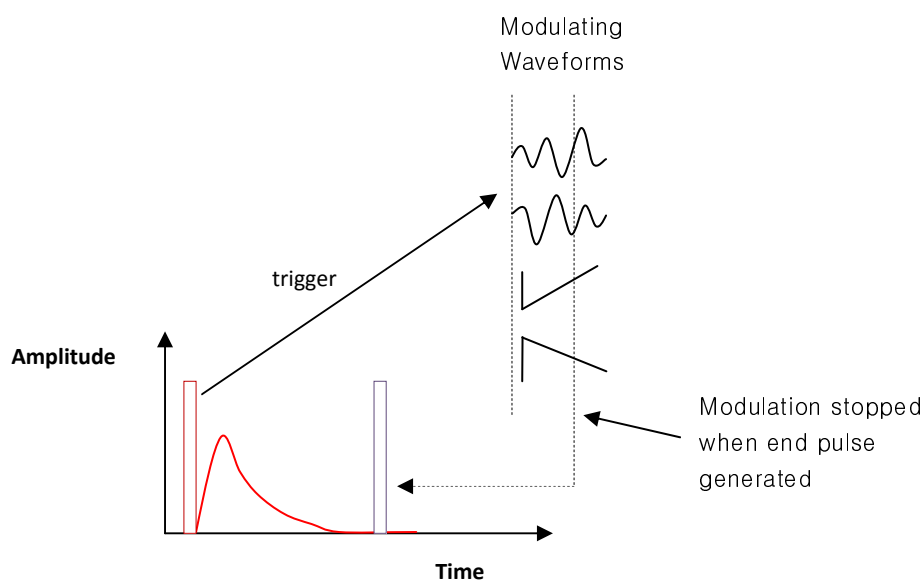


Fig 7

Notice that the slow/fast continuous sine wave modulations waveforms are not included above (even though they are available via the control modulation selector switch) because they are not ‘one-shot’ waveforms but rather are continuous waveforms that are not affected by the start/end pulses generated by the attack/decay detector.

From all this, we see its important for the Hi-FLi to be able to correctly identify the beginning and end of the amplitude envelope of a signal fed into it for proper triggering of the one-shot modulation waveforms.

To get around the problem of multiple ‘triggering’ when a chord is struck on the guitar the attack/decay detector circuit has two sensitivity settings via the sensitivity switch. The ‘solo’ setting is the most sensitive and is designed for single string plucking. In the ‘strum’ setting the attack/decay detector is less sensitive and so it only responds to the larger initial attack peak of a strummed chord and ignores the other smaller peaks in the envelope. In this way it correctly only generates a single start pulse.

However the above does depend on the amount of preamp gain set in the pedal unit, the guitar pickup volume settings and on playing style (on how hard the strings are plucked etc). So in practice you can play chords in ‘solo’ mode and single strings in ‘strum’ mode and still avoid multiple triggering of attack detection pulses, but it requires careful playing. We will discuss how to properly set up the Hi FLi gain settings later.

The Phase Filter Explained

Central to the unique sound shaping of the Synthi Hi-FLi is the Phase Filter. In fact the Phase Filter is a collection of 3 main circuits. The first is the so called vibrato/transient generator which is a function generator producing the 6 modulating waveforms mentioned earlier; slow/fast sine; increasing/decreasing amplitude sine and up/down ramps. The second part consists of a collection of 12 identical phase-shifting filters each of which has the property that it allows waves of all frequencies through without attenuation, but instead phase shifts them by an amount that depends on the wave’s frequency. Hence the name ‘phase filter’ rather than ‘frequency filter’ as one might call more conventional filters like low-pass, band-pas or high-pass types. Each of the 12 filters can produce a maximum phase shift of up to 180 degrees. So if all 12 are used the total maximum shift is $12 \times 180 = 2160$ degrees.

Crucially the amount of phase shift can also be controlled by a voltage. This brings us to the final part of the Phase Filter, the so called ‘bow-string’ drivers. The bow string refers to a collection of 20 diodes strung together end to end and linked to each filter. By applying the modulation waveforms to this string, the amount of phase shifting can be varied.

Since there are 12 such filters this makes a total 240 diodes (!) which is a pretty unique design. The different sound treatments like Vibrato, Phasing etc are achieved by connecting these 12 filters up in different ways and also creating different signal paths through them. So let’s consider each of the 6 main treatments Vibrato, Phasing 1, Phasing 2, Waa, Waw and Meow in turn.

To do this lets represent the 12 filters and the switches that control how they are inter-connected and which also control the signal flow by the following diagram:

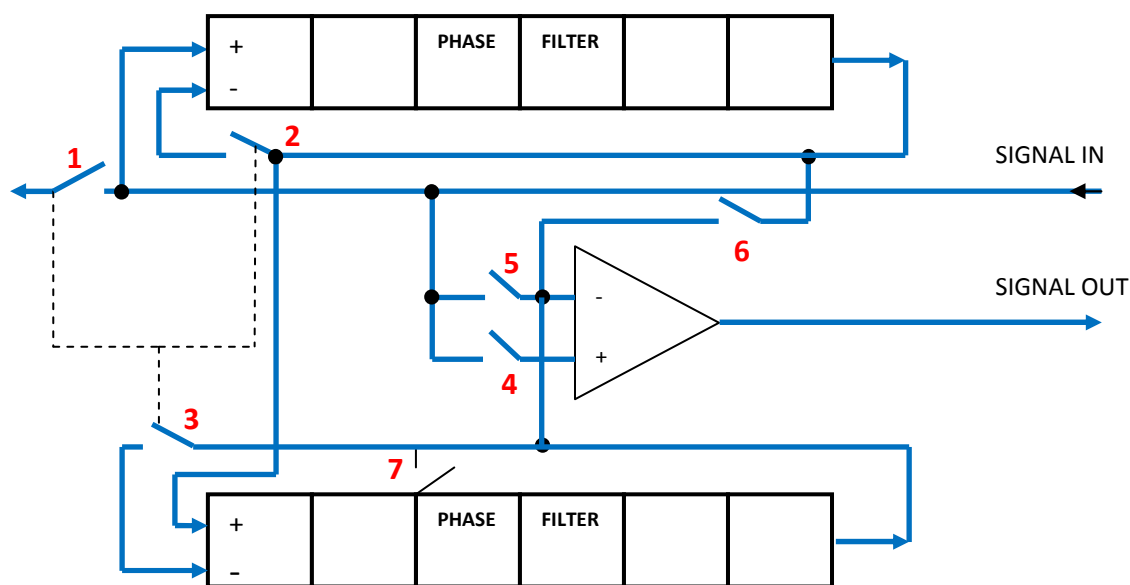


Fig.7 the Phase Filter signal structure

In Fig 7, the 12 phase filters (each filter represented as a square) are arranged in two blocks of 6 filters daisy chained together. The blue line shows the audio path. Notice there are 7 switches labelled 1–7 that can be open or closed and

so create different ‘topologies’. It is these different connections that produce the different treatments. These switches in reality are transistors that can be turned on or off and so act as signal switches.

What is not shown in Fig. 7, but which is implied, is that each of the 12 filters has a modulation voltage controlling the amount of phasing. The modulation source is one of the six available waveforms discussed earlier (slow/fast sine; increasing/decreasing sine; up/down ramps.) These are fed via the ‘bow strings’ (a string of 20 diodes per filter) mentioned above.

The 6 different treatments, Vibrato, Phasing 1, Phasing 2, Waa, Wawa and Meow correspond to the following configurations of the 7 switches:–

TREATMENT	Switches 1,2,3	Switch 4	Switch 5	Switch 6	Switch 7
Vibrato					
Phasing 1			on		
Phasing 2		on			
Waa	on				on
Waw	on			on	
Meow	on			on	

Table 1

Table 1 shows the switch configurations (a blank means the switch is off). From looking at the table, it might appear that there is no difference between the ‘Waw’ and ‘Meow’ effects but in fact there is and we will discuss this below.

Let’s study each treatment in turn.

Vibrato

This is the simplest treatment. Here all the switches are open so the input signal goes once through all 12 filters in turn and back out again. Any variation of phase via modulation waveforms applied to each filter thus modulates the frequency of the signal, i.e. we have frequency modulation. For example imagine a sine wave (e.g. the fast Sine modulation) is used to modulate the

phase shift in each filter. Then we can visualize why the result is a vibrato effect:–

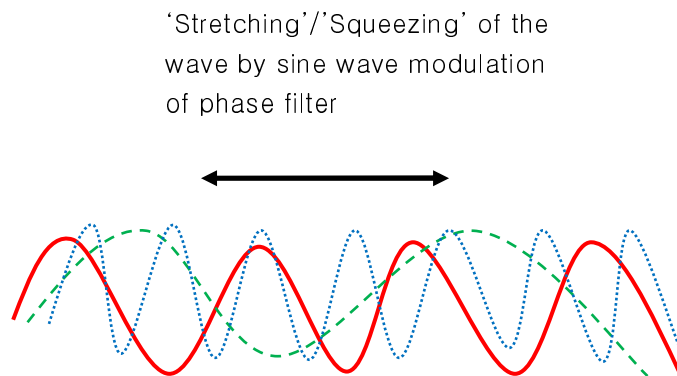


Fig. 8

In Fig. 8 the red wave represent say a particular frequency harmonic in an input signal fed into the filter. By applying the Sine wave modulation to each of the 12 filters, the wave is periodically ‘stretched’ or ‘compressed’ and so its frequency is decreased or increased accordingly. This is just vibrato!

Of course this effect, or rather the amount of stretching is frequency dependent. So higher frequency harmonics may have different amount of vibrato compared to lower frequencies.

In fact you can hear this difference when you set up the Hi-FLi to produce a vibrato effect. We’ll talk about this a little later when discussing the setting up and using the Hi-FLi.

If instead of using a Sine wave to modulate the amount of phase shifting, we were to use say a ramp wave a ‘swept-vibrato’ effect would be heard rather than a repeating vibrato.

Phasing 1

This is very similar to the vibrato routing, but in addition switch 5 is on. This means that the phase shifted signal is mixed with the original signal. As the phase shifting approaches 180 degrees, you therefore get perfect cancelation of a given frequency harmonic. Since each of the 12 filters can at most

produce 180 degrees of phase shifting, the output signal goes to zero 6 times as we sweep the full phase shift range of the 12 filters.

This is illustrated in the following graphic

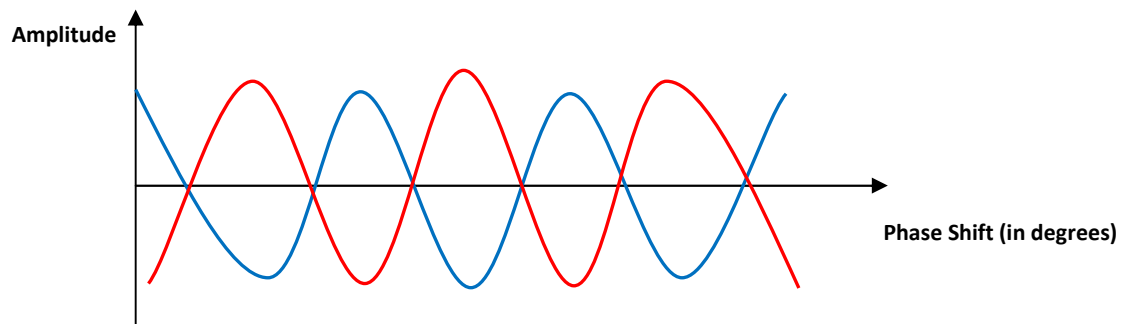


Fig. 9

In Fig. 9 the red wave represent the original input to the filter and the blue the phase shifted wave. When they are mixed together you get perfect cancellation at 180, 540, 900, 1260, 1620, 1980 degrees, i.e. 6 times.

You can actually check this out yourself by operating the Frequency Shift slider with the treatment switch is set to 'Phasing 1'. As you move the slider from 0 to 10 you can hear the output signal increasing and decreasing 6 times as the slider reaches 10. Now if you apply e.g. a slow sine control modulation you will hear the classic 'wooshing' sound of a phaser as the phase shifting is periodically swept by the sine modulation.

Phasing 2

This is quite similar to Phasing 1 except now its switch 4 that is on instead of switch 5. This means the phase shifted signal and the original single are subtracted rather than added together. But again as in Phasing 1, the output will go to zero 6 times as the Frequency Shift slider is moved from 0 to 10. However the sound is different than Phasing 1 as you will discover. (Remember the amount of phase shifting produced by the filter is frequency dependent, so it differs between one harmonic and another!)

Waa

In this setting, switches 1, 2, 3 and 7 are all on. The former means that there is now feedback of the signal output occurring in both the upper and lower filter blocks. Moreover some of the original signal input is actually fed into the phase shift control of all 12 filters (this is indicated by the left pointing arrow near switch 1 in Fig 7). Finally with switch 7 also on, the signal is by-passing that last 4 sections of the lower filter block. The presence of feedback around the phase filter makes them act like a band-pass filter with a single peak occurring because switch 7 is closed. The frequency where the peak is located moves as the phase shift produced by the filter is changed. So if we fed a signal into the filter with the Waa setting and swept through the phase shift by moving the Frequency Shift slider from 0 through 10 we would hear the familiar 'Waa' effect as the frequency of the peak is swept to higher values. Modulating the phase shift by applying control modulations such as slow/fast sine etc will give distinct and powerful 'auto-Waa' effects.

Waw

For this effect, switches 1, 2, 3 and 6 are on. Now all 12 filters are used and there is feedback occurring on the upper and lower blocks of 6. Because switch 6 is on there is mixing between the output signal from the upper 6 filters with the signal that has gone through all 12 filters. As mentioned earlier because of feedback the phase filters act as band pass filters with 3 resonant peaks appearing in the upper and lower filter blocks respectively. As each filter is swept by applying e.g. a slow sine control modulation, the frequency of the 6 resonant peaks themselves move with the varying Sine modulation. The result is an extremely rich effect arising from the combination of phasing as in 'Phasing 1' but with simultaneous resonant filter sweeps as in 'Waa'. It is a unique effect for the Hi-FLi.

Meow

From Table 1 it seems there is no difference between 'Waw' and 'Meow' effects as regards the various switch settings. This is correct but what is not shown in Fig. 7/Table 1 is that in the Meow setting the control modulation applied to the lower block of 6 filters is inverted with respect to the upper block of 6. In 'Waw' the control modulation is the same for all 12 filters (i.e. it's the same for upper/lower blocks). The simple difference of inverting the control modulation applied to the lower block has a very striking effect on the sounds produced! If we imagine again applying a sine wave control modulation then

the 3 resonant peaks of the upper block will now move in opposite directions to the 3 peaks of the lower block. In the case of 'Waw' all 6 peaks move in unison back and forth with sine modulation. Now the two sets of 3 peaks always move in opposite directions. This can lead to amazing vocal/speech like phonetic sounds as the peaks cross! Again this is very likely a unique effect to the Hi-FLi.

So much for the 'theory' behind two very important parts of the Hi-FLi. Let's now move on to discuss its use in practice!

The Hi-FLi in Practice

Setting the sensitivity level of the Hi-FLi

Audio input into the Hi-FLi is via the 6.3mm mono jack input on the side of the right pedal. The pedal unit connects to the main unit via the umbilical and 8 pin plug. The main power to the Hi-FLi is via standard IEC connector. This connector has an integral fuse and 110V/220V selector chosen via a small screw-driver operated switch.

To avoid potential serious damage, please make sure you have chosen the correct voltage setting for the mains power used in your country!

Like the original 1970's units, there is no mains on/off switch in the Hi-FLi. The unit powers up when the mains is operated. You should see the two red indicator leds glow on power up.

Audio out from the unit is via standard 6.3mm mono jack on the rear panel. Connect this to a suitable amplifier.

The first thing to do is to set the input sensitivity switch to 'solo', all 4 left hand sliders to zero, the central mix slider to max (10), fall time knob to slow (fully anti-clockwise). On the right hand side of the control panel, set modulation speed to 5, ramp time to 5, modulation depth to 10 and frequency shift at about 3. Choose the increasing sine control modulation waveform  and 'vibrato' as the filter treatment. Set the Growl switch to its centre (off) position. Make sure all the pedal voltage switches (under each of the 8 effect sliders)